JC TECH KNOWLEDGE HUB -- TECHNICAL SPECIFICATION SERIES

Engineering Tools

High-Precision Design Calculators, Analytical Utilities & Material Database

1. Verified Analytical Utilities for Accelerated Engineering

Spreadsheet-based engineering calculations are notoriously vulnerable to formula errors, broken unit conversions, and unvalidated assumptions. JC Tech Engineering Tools is a cloud-based suite of verified, deterministic calculation utilities engineered to deliver instantaneous design validation for mechanical, structural, and thermal engineers. Every tool is verified against Roark's Formulas for Stress and Strain, Shigley's Mechanical Engineering Design, and governing ASME/ISO/SAE codes, providing engineers with confidence during design conception and review.

2. Core Suite Calculation Modules

- **Beam Bending, Shear & Deflection Suite:** Computes exact bending moments, shear forces, slope, and deflection profiles for simply supported, cantilever, overhanging, and fixed-fixed beams under point, uniformly distributed (UDL), and trapezoidal load conditions.
- **Bolt Preload & Torque-Tension Calculator:** Calculates required tightening torque (T = K * D * F) and induced clamp load per ISO 898-1 and VDI 2230 standards, incorporating nut friction coefficients, thread pitch, and fastener proof stress limits.
- **Tolerance Stack-Up Analysis Engine:** Performs simultaneous 1D/2D Worst-Case Arithmetic analysis and Root-Sum-of-Squares (RSS) statistical tolerance modeling to predict assembly fit, clearance gaps, and expected production scrap rates.
- **Thin & Thick-Walled Pressure Vessel Sizer:** Determines hoop (circumferential) and longitudinal stresses in cylindrical and spherical vessels per ASME Boiler & Pressure Vessel Code (BPVC Section VIII Div 1), including joint efficiency and corrosion allowances.

3. Engineering Material Benchmark Matrix (Standard Properties)

Engineering Material	Yield Strength (MPa)	Tensile UTS (MPa)	Young's Modulus (GPa)	Density (g/cm3)	Thermal Coeff. (1e
Structural Steel AISI 1018	370 MPa	440 MPa	205 GPa	7.87 g/cm3	11.5 um/m-K
Alloy Steel AISI 4140 (Q&T)	655 MPa	850 MPa	210 GPa	7.85 g/cm3	12.2 um/m-K
Aluminum 6061-T6	276 MPa	310 MPa	68.9 GPa	2.70 g/cm3	23.6 um/m-K
Stainless Steel 304 (Austenitic)	215 MPa	505 MPa	193 GPa	8.00 g/cm3	17.3 um/m-K
Titanium Ti-6Al-4V (Grade 5)	880 MPa	950 MPa	114 GPa	4.43 g/cm3	8.6 um/m-K
PEEK Polymer (Polyetheretherketone)	100 MPa	110 MPa	4.1 GPa	1.32 g/cm3	45.0 um/m-K

4. Mathematical Formulation Deep Dive

Calculator Module	Governing Mathematical Formulation	Key Boundary Limits & Assumptions
Cantilever Beam (Point Load at Tip)	Deflection_max = (P * L^3) / (3 * E * I) Moment_max = P * L	Linear elasticity, small deflections (y_max < L/10), isotropic homogeneous material
Simply Supported Beam (Center Load)	Deflection_max = (P * L^3) / (48 * E * I) Moment_max = (P * L) / 4	Plane sections remain plane; ignores shear deformation (Euler-Bernoulli beam theory)
Bolt Torque-Tension Relation	T = K * D * F_preload F_preload = 0.75 * A_tensile * S_proof	Nut factor K accounts for thread & head friction (K=0.20 dry, K=0.15 lightly oiled)
Tolerance Stack (RSS Method)	T_assembly = sqrt(sum [(C_i * T_i)^2]) Z = (USL - Mean) / sigma	Assumes independent normal distribution of individual component machining tolerances
Thin-Walled Cylindrical Vessel	sigma_hoop = (p * r) / t sigma_axial = (p * r) / (2 * t)	Valid only when radius-to-thickness ratio r/t >= 10; uniform internal gauge pressure

5. Digital Tools Operational Ecosystem

- **Instant Unit System Switching:** Toggle seamlessly between SI Metric (mm, N, MPa, deg C) and US Imperial (in, lbf, psi, deg F) with automated decimal precision.
- **Automated PDF Calculation Reports:** Generate client-ready calculation memorandums complete with mathematical derivations, input parameters, and pass/fail safety factor checks.
- **REST API Integration for CAD/ERP:** Embed JC Tech calculation endpoints directly into custom SolidWorks macros, internal web portals, or automated PLM workflows.

6. Suite Performance & Adoption Statistics

24+	100% Code	0.00% Error	Free Access
Specialized Calculators	Grounded in ASME & ISO	Validated vs FEA Meshes	No Credit Card or Login

Launch JC Tech Engineering Tools in Your Browser

Access our complete calculator library online. No installation required. Fully responsive on desktop, tablet, and mobile shop-floor devices.